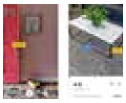

Google kills Measure
Measure, which allowed people to measure things in the real world, has now been discountinued. **http://dgit.in/Jul21-61**

Twitter NFTs with a twist
Twitter users have to reply to the Tweet inorder to get a chance of winning them. **http://dgit.in/Jul21-62**

Pop in BTRFS

Install Pop!_OS on BTRFS with Backups and Data Deduplication

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BTRFS

BTRFS, often pronounced as Butter-FS is a modern copy-on-write filesystem for Linux systems. It brings to the table features like transparent compression, snapshots, subvolumes, multi device support, deduplication, incremental backups, and more. Chris Mason, the principal Btrfs author, stated that its goal was "to let [Linux] scale for the storage that will be available. Scaling is not just about addressing the storage but also means being able to administer and to manage it with a clean interface that lets people see what's being used and makes it more reliable".

POP!_OS

Pop!_OS is a Linux distribution maintained by System 76. It is derived from Linux Ubuntu, and adds many tweaks and quality of life improvements over Ubuntu. Particularly, it works like a charm on hardware shipped by System 76, and is also the only distribution which lets me use the NVIDIA dGPU on my laptop. Find out more about Pop!_OS on the official website at https://pop.system76.com. The guide is for pop 20.10, and if you user an older / newer version, "your experience may wary"

THE INGREDIENTS

For this particular article, we consider a computer with two storage drives of roughly equal performance. This could be two HDDs, or two SSDs, or even something else, as long as they are in the same performance class. You could apply this guide to a setup with a very fast drive and a much slower drive and it should still work, but the overall performance of the resulting setup would be abysmal. You could, however, also apply the guide to three or more drives of comparable speed.

We will use BTRFS' ability to pool disks together to combine the storage of both the drives, and present it to the OS as one large filesystem. As this setup is not supported by Pop!_OS' installer by default, we will need to install Pop on a configuration it is compatible with, and then move things around to be in line with our goals.

Note that much of the trickery involved is in getting the right subvolumes in place. If you already have a btrfs setup, you can simply add another disk to the pool without jumping through hoops.

The guide also assumes you are running a UEFI computer. Any recent computer should support UEFI, as do many hypervisors.

TL;DR

In very short, we take the following steps:

1. Boot Pop!_OS installer, enter the locale settings
2. Open gparted, partition one of your disks to have ESP, Recovery, Swap, and a BTRFS partition. On the other disk, create a single large partition without a filesystem
3. Install Pop using the BTRFS partition on the first disk as root, and the BTRFS partition on the second disk as home. When the installer asks to reboot, ***do not reboot!***
4. Mount the root partition somewhere, create a BTRFS subvolume named `@` for root, and move everything into it. Also create a subvolume for home named `@home`.
5. Add the large btrfs partition into the mountpoint, and balance your data across both drives
6. Bind mount `/dev, /dev/pts, /proc, /sys, and /run` into your mountpoint, and `chroot` into it. Update your `fstab` by hand. You can use `kernelstub` command line utility to add rootflags and update bootloader configuration.
7. Automount your drives via `fstab`, and check that there are no errors. Optionally add a timeout to the loader selection screen at `/boot/efi/loader/loader.conf`

THE DETAILS

Pop Installer

Boot off your Pop install media. Before anything else, the installer asks you to confirm your locale information : language, keyboard layout, etc. Once you go through this process, it will present you the option to install or try the demo environment. At this point, we want to click on the Try Demo button, which will close the installer, and leave us in a full fledged Pop!_OS environment.

Disk Preparation

Use `lsblk` to find out what the names of your drives are. For HDDs, it should be something like `sda` and `sdb`, and for SSDs, it should be something like `nvme0n1` and `nvme1n1`. Remember which are the drives you want to dedicate to your install. If you find yourself confused with which drives you want to wipe, consider unplugging any drives you do not want to modify. In this guide, the first drive is `/dev/sda` of size 50